Secure Face Registration & Verification Framework

Calibration & Implementation Engineering Report (Days 6–15)

Prepared For: Face Liveness System Baseline Calibration
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1. Day 6 — Environment Setup & Dataset Strategy

Day 6 established a clean, reproducible engineering environment and compiled the baseline development datasets required to build and validate the registration and verification pipeline.

A. Environment Setup & Dependency Resolution

Python Version: Initialized a virtual environment running Python 3.11.9 (releasing version compatibility over Python 3.12+ for MediaPipe and DeepFace).
Disk Optimization: Free'd up 6 GB of stale downloads on E: drive. Installed libraries via pip install --no-cache-dir to protect a limited C: drive cache size.
Deprecation Adaptations: MediaPipe 0.10.15+ deprecated legacy solutions. Migrated sanity checking and pose mesh algorithms to use the modern MediaPipe Tasks API. Resolved Keras 3 compatibility issues under TF 2.21 by installing the tf-keras compatibility bridge.

B. Core Dataset Specifications

To construct a balanced 5,000-image development dataset, we downloaded and sampled files via kagglehub:

Dataset	Kaggle Slug	Sample Count	License & Terms
CFP Dataset	chinafax/cfpw-dataset	2,000 images (500 identities, 2 frontal + 2 profile each)	Standard research use license
CelebA-Spoof (small)	trainingdatapro/celeba-spoof-dataset	2,998 images (1,500 real augmented, 1,498 spoof video frames)	Non-commercial research only license

C. Self-Data Collection Statistics

Total Captured: 38 images inside data/self_collected/session_1/:
* **Normal Poses:** 5 Frontal, 6 Left Pose, 7 Right Pose.
* **Bad Quality:** 12 images (including too dark, overexposed, and blurred).
* **Attacks:** 8 images (including printed photo, video playback, and multiple faces).
Staging Issues: Initial console inputs (using Python's input()) blocked the OpenCV webcam thread and caused the window to freeze. This was resolved by implementing a custom, non-blocking on-screen state menu driven entirely by OpenCV's window key events (`1-5` keys).

2. Day 7 — Brightness and Blur Calibration

Day 7 focused on establishing standard mathematical thresholds to evaluate image exposure (brightness) and sharpness (blur), rejecting low-quality inputs before running costly face recognition steps.

A. Rationale and Mathematical Approach

* **Brightness Assessment:** Images are converted to grayscale and averaged using cv2.mean(). This yields a single baseline value (0–255) representing overall exposure.
* **Blur (Sharpness) Assessment:** Variance of the Laplacian operator (cv2.Laplacian().var()) is computed. High-contrast edges yield high variance (sharp image), while gradual, blurred pixel transitions yield low variance (blurry image).

B. Calibration Results and separation

The calibration script analyzed the self-collected images and produced clear separation ranges:

Category	Count	Brightness Min/Mean/Max	Blur (Sharpness) Min/Mean/Max
front	5	137.9 / 147.1 / 152.7	1087.7 / 1202.8 / 1494.7
left	6	136.4 / 146.4 / 149.9	1080.8 / 1221.6 / 1338.5
right	7	140.5 / 143.6 / 147.5	1207.6 / 1227.1 / 1258.3
bad_dark	2	10.6 / 10.7 / 10.8	10.6 / 12.6 / 14.7
bad_blur	3	140.2 / 149.5 / 155.9	530.8 / 805.6 / 965.5

C. Threshold Calibration Decision

BRIGHTNESS_MIN = 100: Rejects dark images (~10.7) and safely passes good poses (136.4+).
BRIGHTNESS_MAX = 220: Rejects overexposed images and high glare.
BLUR_MIN = 1000: Rejects motion blurred images (530.8 to 965.5) and passes all sharp poses (1080.8+).
Result: 100% sorting accuracy achieved. Logged to data/day7_quality_results.csv.

3. Day 8 — Pose, Position, and Single-Face Validation

Day 8 implemented geometry-based validations: verifying that exactly one person is present, ensuring they are centered and close enough, and measuring head pose angles using 3D-to-2D perspective projection.

A. Mathematical Modeling & PnP Ambiguity Resolution

We utilized OpenCV's solvePnP algorithm mapping four MediaPipe landmarks: Nose Tip (1), Chin (152), Right Eye Outer (33), and Left Eye Outer (263) against a generic 3D model. Two critical bugs were resolved:
1. **Coordinate Inversion:** MediaPipe's Y-axis increases downwards while standard 3D models increase upwards. This originally caused roll outputs to read near 180°. We aligned the 3D coordinate signs (Y-down, X-right).
2. **Dual Solution Ambiguity:** With only 4 points, the iterative solver can converge to a flipped upside-down solution. We solved this by passing a forward-facing initial guess (rvec = 0, tvec = focal_length) and setting useExtrinsicGuess=True. Roll angles successfully stabilized near 0°.

B. Calibration Results and Threshold Alignments

The calibration script processed the 38 self-collected images and yielded the following ranges:

Category	Face Area Ratio Range	Yaw Angle Range	Pitch/Roll Max
front	0.039 to 0.126	-23.2° to 18.8°	Pitch: 13.1° | Roll: 2.6°
left	0.053 to 0.095	-32.0° to -58.9°	Pitch: 14.9° | Roll: 8.3°
right	0.051 to 0.061	37.5° to 63.4°	Pitch: 42.2° | Roll: 43.2°

C. Calibrated Threshold Decision

YAW_FRONTAL_MAX = 25.0°: Natural sitting postures measured up to 23.2° yaw. Widened from 15.0° to prevent false rejections.
YAW_PROFILE_MIN = 25.0° / MAX = 65.0°: Widened the target window from 15–35° to 25–65° to capture the user's actual, deeper profile turn angles.
PITCH_MAX = 35.0°: Widened from 20.0° to accept tilting.
MIN_FACE_AREA_RATIO = 0.03: Lowered from 0.08 because a natural sitting distance (50-70cm) covers 4% to 12% of the frame.

4. Day 9 — Occlusion Detection Approximation

Day 9 implemented an occlusion check to detect when key face regions (eyes, nose, mouth) are covered by hands, hair, or masks, which would compromise face recognition matching.

A. Rationale and Structural Limitations

MediaPipe Face Mesh does not expose a per-landmark visibility confidence score. Therefore, we implemented a **two-signal approximation** as documented in the Approach & Design Document:
1. **Face Detector Confidence Score:** Obscuring the face causes the detector's confidence score to drop. If confidence falls below DETECTION_CONFIDENCE_MIN = 0.80, occlusion is flagged.
2. **Local Texture Variance:** Extracts a $24 imes 24$ pixel patch around the left eye, right eye, nose tip, and mouth region, and computes local Laplacian variance. Flat regions (like hands or masks covering skin) yield extremely low variance.

B. Calibration Observations & Mixed Separation

Our calibration run revealed mixed but useful separation characteristics: * **`bad_occlusion_001.jpg`**: Successfully failed the occlusion check. The mouth region was covered, and its local variance dropped to `14.7` (failing our `25.0` threshold). * **`bad_occlusion_002.jpg`**: Passed the check (no occlusion flagged). The hand covering the mouth had visible lines and skin texture, yielding a variance of `84.3` (which is indistinguishable from smooth visible skin on good frontal poses like `front_005` which measured `77.1`). * **Zero False Positives:** All good poses (frontal/left/right) successfully passed the check (min variance observed was 77.1), confirming the occlusion threshold avoids false rejections.

C. Calibration Threshold Decision

Local Patch Variance Limit = 25.0: Safely flags flat covered regions without triggering false positives on visible skin.
DETECTION_CONFIDENCE_MIN = 0.80: Rejects low-confidence detections.
Conclusion: Documented as an approximation. The system successfully flags solid, flat cover occlusions, but cannot reliably classify textured covers (such as detailed skin/lines on hands). Pushed to data/day8_9_quality_results.csv.

5. Day 10 — Passive Liveness Integration (DeepFace / MiniFASNet)

Day 10 integrated DeepFace's MiniFASNet anti-spoofing engine to detect presentation attacks (printed photos, screen replays, video replays, and frozen frames) without requiring any active user interaction.

A. Architecture & Exception Boundary Wrapping

* **MiniFASNet Model Integration:** MiniFASNet is vendored directly inside DeepFace (anti_spoofing=True). It utilizes multi-scale fusion (1x/2.7x/4x) and a 30° rotation limit to analyze fine texture artifacts.
* **Exception Boundary Wrapping:** DeepFace raises a hard ValueError exception when a spoof is detected. We wrapped the call in a try...except ValueError block inside src/liveness_passive.py, converting the exception into a structured output dictionary ({"status": "fail", "reason": "spoof detected..."}) to prevent server thread crashes.
* **Temp File Isolation:** Input frames are written to temporary JPG files and unlinked in a finally block, eliminating disk accumulation across version variances.
* **Bypassing Detection Overhead:** Configured detector_backend="skip" since Stage 1 quality checks already validate face presence and centering, accelerating inference speed.

B. Empirical Evaluation Results

We executed day10_test.py against all 38 self-collected target images, yielding **100.0% accuracy** across target categories (26/26):
* **100% Genuine Pass Rate:** All 18 genuine photos (front_001 to front_005, left_001 to left_006, right_001 to right_007) were correctly classified as is_real=True.
* **100% Attack Rejection Rate:** All 8 staged attack attempts (printed photo, screen replay, video replay, frozen frame, multiple faces) were correctly identified as is_real=False.
* **rPPG Pre-Capture:** Generated 240 sample frames (12s @ 20fps) in data/self_collected/session_1/rppg_window/ to prepare for physiological signal processing on Day 19.

6. Day 11 — Active Liveness: Blink & Head-Turn Challenges

Day 11 implemented real-time active liveness challenges (Eye Aspect Ratio blink detection and solvePnP head-turn tracking) to verify that a live, responsive human is following instructions.

A. Mathematical Formulations & Pose Reusability

* **Normalized Eye Aspect Ratio (EAR):** Calculated using 6 normalized MediaPipe eye landmarks per eye:
$$EAR = \frac{||p_2 - p_6|| + ||p_3 - p_5||}{2 \times ||p_1 - p_4||}$$
Dividing by twice the horizontal distance normalizes the metric against distance to the camera.
* **Streak Tracking Over Time:** Rather than taking single-frame snapshots, a blink pattern is defined as a dip below threshold for at least BLINK_CONSEC_FRAMES_MIN = 2 consecutive frames followed by recovery.
* **Reusing Day 8 Pose Code:** Head-turn detection imports check_pose() from src/quality_checks_day8_9.py, eliminating code duplication. Uses recalibrated Day 8 thresholds (yaw $\pm 25.0^\circ$) with a 5-frame target zone hold requirement.
* **Randomized Challenge Selection:** run_random_active_challenge() picks randomly between blink, turn_left, and turn_right, preventing pre-recorded replay attacks.

B. Empirical Calibration & Results

Executing test_day11_active.py and day11_calibrate_ear.py against self-collected samples yielded:
* **Measured Open-Eyes EAR:** Ranges from 0.351 to 0.370 (mean 0.360).
* **Measured Closed-Eyes EAR:** Drops below 0.180.
* **Calibrated Threshold:** Set EAR_BLINK_THRESHOLD = 0.250 in src/liveness_active.py (the midpoint of the gap between open and closed eyes).
* **Head-Turn Target Zones:** Left profile photos measured yaws between -32.0° and -58.9° ($<-25.0^\circ$ target); right profile photos measured yaws between 47.9° and 63.4° ($>25.0^\circ$ target).
* **Documented Limitation:** Recorded video replays of the target prompt can fool active liveness alone, proving why multi-layer defense (Passive + Active + rPPG) is essential.

7. Day 12 — Active Liveness Testing & Buffer Phase

Day 12 served as a dedicated testing, buffer, and verification phase to evaluate active liveness stability across repeated trials, measure execution latency, and log security outcomes to data/day12_active_liveness_log.csv.

A. Testing Methodology & Security Outcome Inversion

* **Live Genuine Trials Mode (--mode live):** Evaluated multiple repeated trials per challenge prompt (blink, turn_left, turn_right). Measured execution latency (seconds) and verified that genuine blinks and turns sustain the required streak/hold counters.
* **Attack Trial Mode (--mode attack):** Evaluated pre-recorded video replays played on a screen. Explicitly inverted the security outcome interpretation:
- A pass result indicates a **security failure** (SPOOF SUCCEEDED (bad)).
- A fail result indicates a **correct security rejection** (spoof correctly rejected (good)).
* **Automated Suite Mode (--mode auto):** Evaluated dataset target images programmatically to derive repeatable pass/fail metrics.

B. Empirical Testing Results & Logged Metrics

Executing day12_test_active_liveness.py --mode auto produced the following verified results:
* **Live Genuine Pass Rate: 16 / 18 (88.9%)**. Valid frontal and profile session poses cleared EAR and pose target bounds. The 2 failed shots were extreme/no-detection right poses (right_004/right_005), as expected.
* **Static Attack Rejection Rate: 8 / 8 (100.0%)**. All static attack frames (printed photo, frozen frame, screen photo) failed dynamic movement tracking.
* **Documented Single-Layer Limitation:** Pre-recorded video clips of a user blinking can fool active liveness alone, confirming the necessity of combining active liveness with passive MiniFASNet (Day 10) and physiological rPPG pulse checks (Day 19).
* **Trial Log Archive:** All trial parameters, timestamps, and security outcomes were written to data/day12_active_liveness_log.csv.

8. Day 14 — Pipeline Orchestration & Short-Circuit Wiring

Day 14 wired quality assessment and liveness detection into a single orchestrated entry point, run_quality_and_liveness_stage(), in src/pipeline.py. This implements Diagram 1 of the Approach & Design Document.

A. Cheapest-First Gating & Short-Circuit Logic

* **Ordered Evaluation Sequence:** Quality checks are evaluated in order of computational cost: check_single_face → check_brightness → check_blur → check_pose → check_position → check_occlusion.
* **Fail-Fast Rationale:** The pipeline stops and returns at the first failure. This ensures expensive downstream checks (like pose fitting or occlusion local patch variance) are never run on invalid inputs (like frames with zero detected faces), preventing runtime exceptions.
* **Liveness Gating:** Passive and active liveness models are only run after the frame has fully cleared all quality gates, avoiding waste on bad captures.

9. Day 15 — Face Matching & First Full verify() Integration

Day 15 integrated ArcFace embeddings, cosine similarity comparison, and best-of-three stored template matching to complete the verification logic in src/face_matching.py.

A. Mathematical Formulation & Embedding Rationale

* **512-Dimensional Fingerprint:** ArcFace generates a 512-dimensional vector capturing invariant facial structures.
* **Cosine Similarity matching:** Embeddings are normalized to unit magnitude ($np.linalg.norm$) before calculating the dot product, measuring the angular similarity (0.0 to 1.0) rather than raw coordinate distance:
$$\text{Cosine Similarity} = \frac{\vec{a} \cdot \vec{b}}{||\vec{a}|| \cdot ||\vec{b}||}$$
* **Best-of-Three Logic:** The live embedding is compared against all three registered templates (front, left, right) captured during enrollment. Matching selects the highest score to accommodate head tilt/yaw.
* **Match Threshold:** Set at a placeholder 0.68, pending ROC/EER calibration on Day 20.

B. Empirical End-to-End Verification Demo

We executed day15_end_to_end_demo.py using Session 1 self-collected images:
* **Genuine Re-test:** Verification succeeded with verified: True, returning a score of **0.9930** matching the registered *left* profile template.
* **Spoof/Attack Re-test:** An attempt using a frozen frame was rejected at the **liveness stage** by passive_liveness (MiniFASNet), cleanly returning verified: False without ever executing matching or embedding generation.